Arpan Pathak

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Visa: Requires H1B Transfer sponsorship

SUMMARY

Systems Software Engineer with 7+ years architecting high-performance distributed systems, ML-driven platforms, and cloud-native infrastructure at **Amazon**, **Microsoft**, and **Oracle**. Deep expertise in **Rust**, **C++**, **systems programming**, and **machine learning** — building low-latency, high-throughput engines at global scale. Proven track record in eBPF/kernel-level networking, lock-free data structures, AI model deployment (BERT, XGBoost, Deep Learning), and Kubernetes infrastructure. Passionate about GPU-accelerated computing, performance engineering, and pushing the boundaries of what software can do at the hardware-software interface.

EXPERIENCE

Software Developer II — **Microsoft** — Redmond, WA | June 2025 – Present

eBPF & Kernel-Level Networking: Designed and deployed eBPF-based network policies using **Cilium** for Microsoft Purview's sovereign cloud regions. Engineered the control plane to enforce L7 filtering based on geographic traffic origins, and authored custom **Kubernetes Operators** (Kubebuilder) to reconcile compliance states automatically across AKS clusters in France and Germany.

AI-Powered Policy Engine: Built an AI-driven engine that auto-generates Cilium network policies and scans for misconfigurations, reducing manual policy review cycles from 4 hours to near-real-time with zero policy drift across sovereign regions.

C++ to Rust Migration: Spearheaded the migration of the Information Protection service from **C++ to Rust**, shifting memory safety to compile-time. Eliminated use-after-free and buffer-overflow CVEs in the audited path — a critical win for DORA compliance — while maintaining full performance parity.

Infra: **Rust, C++, Cilium/eBPF, Kubernetes (AKS), Kubebuilder, Linux, Azure**

Senior Software Engineer — **Oracle Cloud Infrastructure** — Seattle, WA | Oct 2024 – Mar 2025

High-Performance Database Engine: Architected low-latency internals for a proprietary database engine, optimizing **lock-free data structures** to reduce contention in high-concurrency read-write paths, improving transaction throughput by **45,000 ops/s**.

Infrastructure-as-Code at Scale: Engineered Terraform provider for Oracle Autonomous Database, replacing legacy control plane infrastructure with projected **$1.2M/month cost savings**.

Security & Key Management: Integrated OCI Security Vault across control plane services for secure secret management and automated key rotation protocols.

Infra: **Rust, C++, Go, Terraform, OCI, PostgreSQL internals**

Software Development Engineer II — **Amazon** — Seattle, WA & Hyderabad, India | Oct 2021 – Oct 2024

Real-Time ML Ranking Engine: Architected an **ensemble model pipeline combining XGBoost and Deep Learning** for real-time ads & deal ranking across **FreeVee, Twitch, MiniTV, Prime Video, and Amazon Retail**, driving **5% CTR increase**, 30% higher deal impressions, and **$5.4M monthly revenue growth**.

High-Throughput Distributed Systems: Scaled the Ad Sponsorship Campaign Management platform across 10 geographic regions with a real-time Event Bus & Audit Trail, achieving **100ms latency** while processing millions of ad inventory deals per second.

Conversational AI (BERT): Developed a **BERT-based conversational AI model** and UI that reduced Amazon WorkEvent support ticket triage SLA from 7 days to **2 hours** while handling **5,000 QPS** at peak.

Distributed Data (Serializable Isolation): Designed a uniqueness constraint indexing solution and SDK for a multi-tenant NoSQL datastore using **two-phase commit** for list-type fields, achieving **serializable isolation** at 3K writes/second.

Real-Time Monitoring & Optimization: Reduced over-delivery of deal impressions by ~30% and budget overspend by 25% through real-time monitoring and automated controls.

Spark & Big Data Pipelines: Engineered and fine-tuned **Apache Spark** jobs processing millions of daily events into Amazon Redshift for OLAP, with automated BI dashboards.

Microfrontend Architecture: Led migration to React.js microfrontend architecture, reducing deployment cycles from 5 days to **30 minutes**.

Infra: **Python, Go, Java, Spark, Kafka, DynamoDB, Redshift, SageMaker, AWS**

Software Development Engineer

Razorpay — Bengaluru, India | Aug 2020 – Feb 2021

Built UPI Payment Gateway Aggregator using **Golang** and Protobuf, processing **1M+ hourly transactions**.

Maintained **99.99% uptime** with **150ms P99 latency** under 50,000 TPM peak loads.

Infra: **Go, Kafka, Redis, PostgreSQL**

Software Engineer

Mindfire Solutions — Bhubaneswar, India | Aug 2018 – Feb 2020

- Developed full-stack workflow tools and AR-based product search microservice.
- Built administrative dashboards reducing data retrieval latency from 5 min to 1 min.

Infra: **Java, React, WebSocket, MongoDB**

SKILLS

Category	Technologies
Languages	Rust, Kotlin, Java, Go, Python, TypeScript, C++, C#, Shell Script
Distributed Systems & Cloud	AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS, Bicep), Kubernetes (EKS, AKS), Docker, Terraform, AWS CDK, Apache Kafka, Apache Spark, Apache Flink, Kubebuilder, Istio, Cilium, Kubernetes Gateway API
Distributed Data & Storage	DynamoDB, PostgreSQL, Redis, MongoDB, Amazon Redshift, Amazon S3
Frontend Engineering	React.JS, JavaScript, Angular, HTML 5, SCSS
Observability	AWS CloudWatch, Prometheus, Grafana, Azure Monitor
Machine Learning & AI	TensorFlow, PyTorch, Scikit-learn, XGBoost, Transformer Models (BERT), Recommendation Systems, Deep Learning, Neural Networks, MLOps, Natural Language Processing (a.k.a Computer Linguistics), Computer Vision, Audio Signal Processing
Core CS Fundamentals	Data Structures & Algorithms, Distributed Systems (consensus, replication, sharding), Concurrency, System Design, Networking (TCP/IP, UDP, TLS, DNS, HTTP 3, load balancing)
Systems Programming	Rust, C++, C, C#, eBPF, Cilium, Lock-Free Data Structures, Memory Safety, Concurrency

EDUCATION

Bachelor of Technology — Computer Science & Engineering

RCC Institute of Information Technology — Kolkata, India | 2018